Retail sales in Databricks: the course SQL overstated orders by 83%

Databricks SQL · Unity Catalog · Parth Dhawan · October 2026

4 tables · 4,494 order lines · 6 course queries · 5 corrected queries · 5 Databricks charts

Two counting mistakes, then one clear customer story



+83%

orders overstated

4,494 product lines were counted as orders. There are 2,453 real orders.



+22%

sales overstated

€1.83M included €303K of tax and €23K of shipping. Net sales are €1.50M.

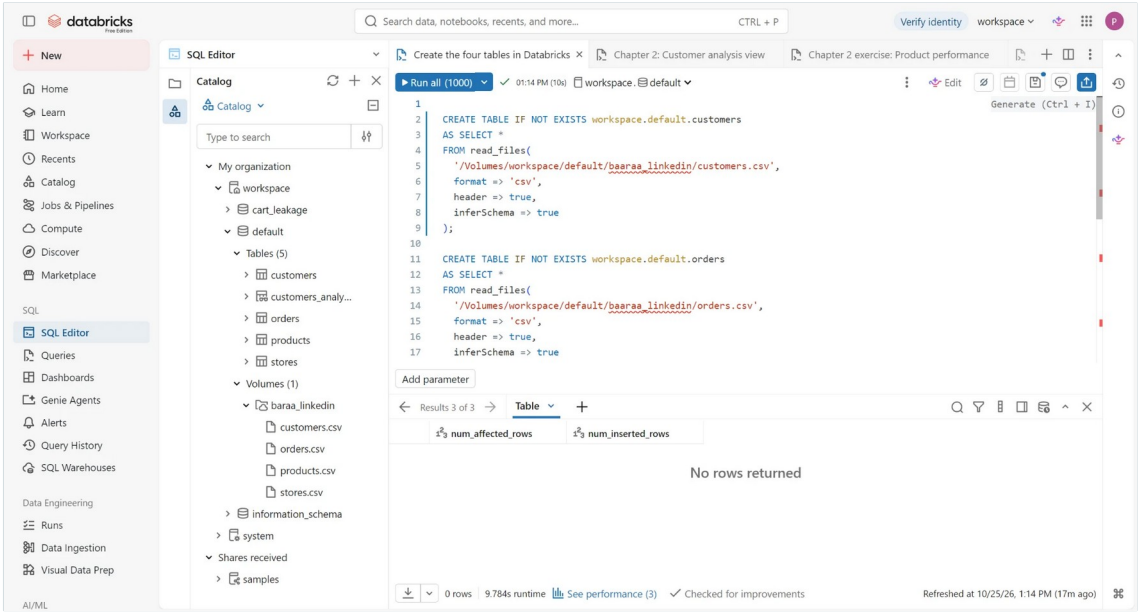


60%

from the top 20%

of net sales, after the fix: 160 of 800 customers bring €900,673.

Built end to end in Databricks SQL



Volume
4 CSVs uploaded to Unity Catalog

Delta tables
Loaded with read_files and schema inference

SQL editor
6 course queries and 5 charts

Corrected views
Every number checked against the raw tables

Data: 800 customers, 4,494 order lines, 250 products and 20 stores across 10 European countries, 2022 to 2025.

Spot it: rows 1 and 2 are one order, counted twice

	order_id	total_amount	customer_id	product_id	store_id	order_at	o
1	O0000001	416.55	C000332	P000004	S00014	2025-09-08T13:28:50.000+00:00	deliv
2	O0000002	123.33	C000332	P000028	S00014	2025-09-08T13:28:50.000+00:00	deliv
3	O0000003	212.41	C000635	P000245	S00004	2025-10-12T10:17:05.000+00:00	deliv

Same customer

C000332

Same second

13:28:50

Same store

S00014

One order number

ES-202512-28364

4,494 rows ÷ 2,453 order numbers = 1.83 product lines per order. So COUNT(order_id) counts products in baskets, not orders. The fix is COUNT(DISTINCT order_number).

"Sales" included tax and shipping



€1.83M

what the course called sales (total_amount)

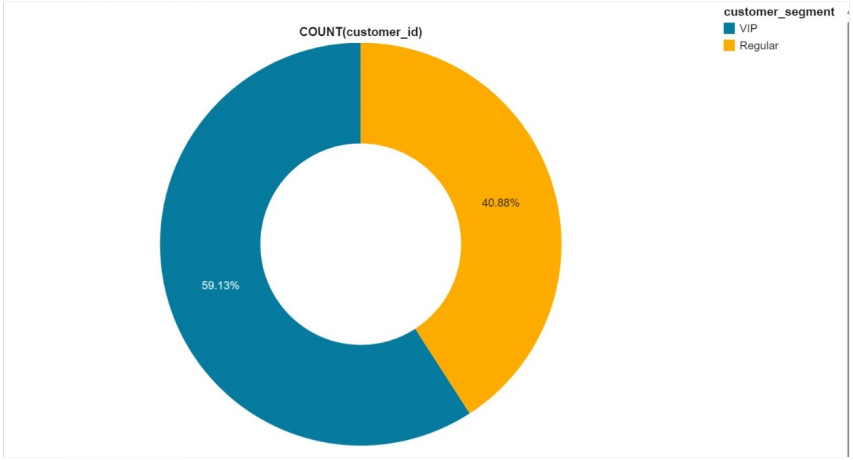
€1.50M

net sales: line subtotal minus discount

0 of 4,494

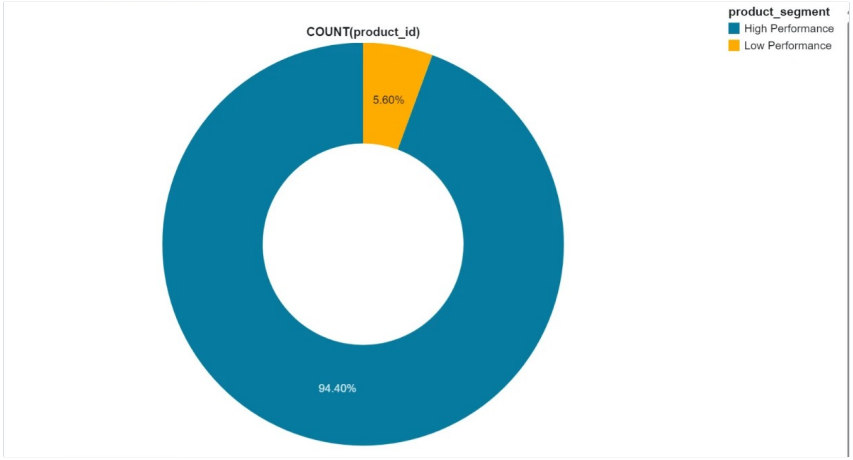
rows where net + tax + shipping ≠ total

A flat €1,000 rule makes almost everyone "top"



59% of customers are "VIP"

Course view: spent over €1,000 = VIP



94% of products are "High Performance"

Course exercise: €1,000 or more = High

Fix: rank instead of a fixed amount. Top 20% = 160 customers and 50 products, a segment that actually separates.

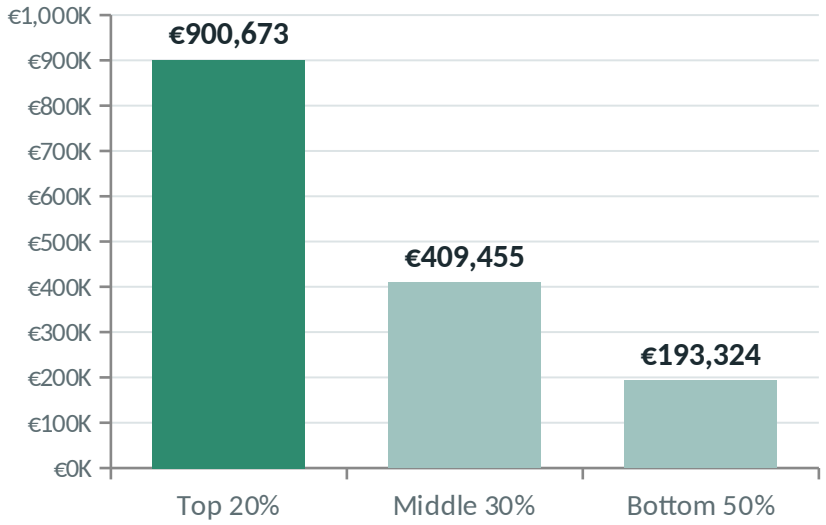
Four fixes, each checked against the raw tables

Course SQL	Problem	My fix	Result
`COUNT(order_id)`	Counts product lines	`COUNT(DISTINCT order_number)`	**2,453 orders**
`SUM(total_amount)`	Adds tax and shipping	`line_subtotal − discount`	**€1.50M net sales**
> €1,000 = VIP	59% of customers are VIP	NTILE ranking: top 20%	**160 customers**
`GROUP BY order_at`	One row per order time	One row per customer	**800 rows, 1 each**

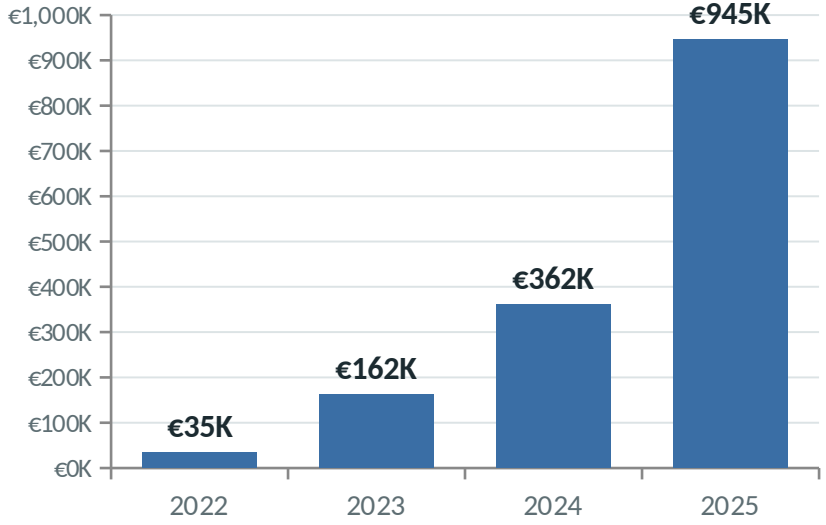
The course versions stay unchanged in the repo next to my versions, with the expected result written under every corrected query.

After the fix: 160 customers bring 60% of sales

Net sales by customer segment

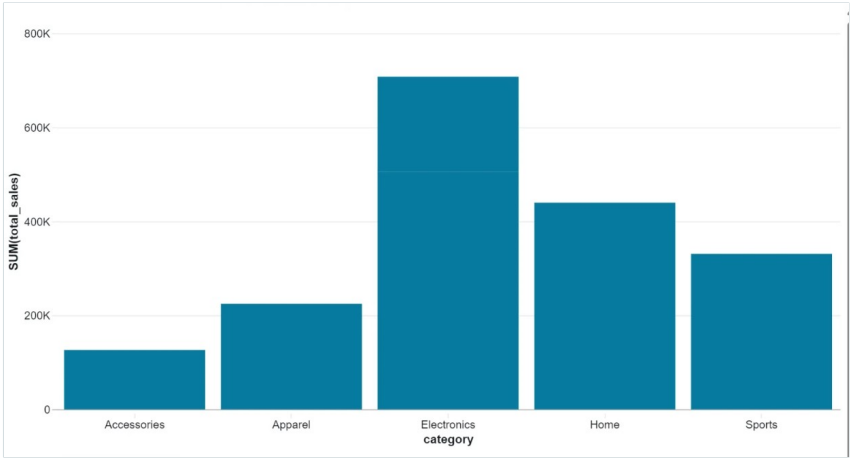


Net sales by year (2022 starts in Feb)

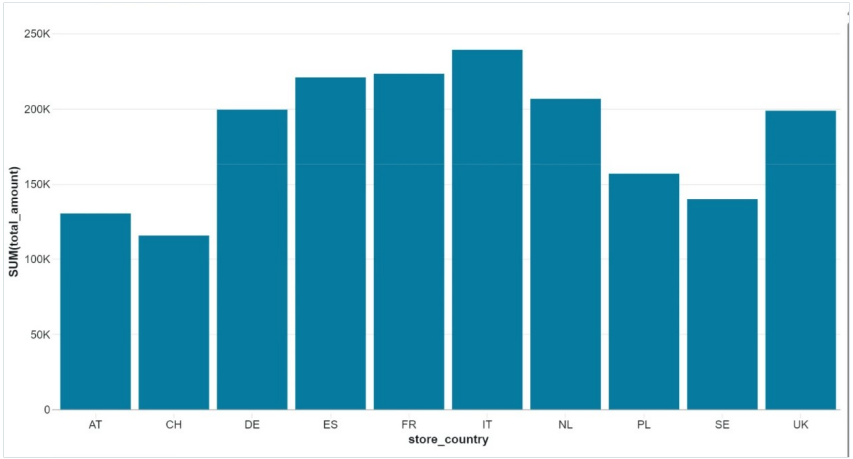


€612.90 average order · **25.8%** gross margin · **63%** of four years of sales came in 2025

Where sales come from



Electronics leads with 39% of sales



Italy sells most, Switzerland least

Physical stores bring 51% of sales and online 49%. These charts come from my Databricks SQL editor on the course queries, so amounts include tax; on net sales the shares and rankings hold.

Next steps



1 Protect the top 160 customers with a retention offer: they hold 60% of net sales.



2 Rebuild the dashboard counters on the corrected view: distinct orders and net sales.



3 Fix the source: order-number dates disagree with order_at on 4,416 lines, and 377 sales predate the product launch.



4 Run the data-check query before every refresh, so a counting mistake never reaches a report.

Happy to walk you through the SQL

Course SQL and my corrected SQL sit side by side in the repo, with expected results under every query.


github.com/parthdhawan921-ops/databricks-sql-analytics


parthdhawan921-ops.github.io/Parth_Personal/#dbx


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